

Deep Neural Network-based Optimization Framework for Safety Evacuation Route during Toxic Gas Leak Incidents

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ARTICLE INFO

Keywords:

Evacuation
Toxic gas leak
Computational fluid dynamics
Surrogate model
Variational autoencoder

ABSTRACT

Evacuation planning is important for reducing casualties in toxic gas leak incidents. However, most evacuation plans are too qualitative to be applied to unexpected practical situations. Here, we suggest an evacuation route proposal system based on a quantitative risk evaluation that provides the **safest route for individual evacuees by predicting dynamic gas dispersion** with a high accuracy and short calculation time. Detailed evacuation scenarios, including **weather conditions, leak intensity, and evacuee information**, were considered. The proposed system evaluates the quantitative risk in the affected area using a **deep neural network surrogate model to determine optimal evacuation routes by integer programming**. The surrogate model was trained using data from **computational fluid dynamics simulations**. A variational autoencoder was applied to extract the geometric features of the affected area. **The predicted risk was combined with linearized integer programming to determine the optimal path in a predefined road network**. A leak scenario of an ammonia gas pipeline in a petrochemical complex was used for the case study. **The results show that the developed model offers the safest route within a few seconds with minimum risk**. The developed model was applied to a sensitivity analysis to determine variable influences and safe shelter locations.

1. Introduction

Chemical plants house hazardous liquids and gasses that may cause fires, explosions, or poisoning upon leaking into the atmosphere. In particular, leakage of toxic chemicals may lead to extensive damage and numerous casualties in a short period following dispersion if the chemical plant site is located near residential areas. One well-known case is the methyl isocyanate gas leak incident at the Union Carbide pesticide plant in Bhopal, Madhya Pradesh, India. Approximately 40 tons of gas leakage killed more than 3800 people immediately and caused at least 15,000 premature deaths over the subsequent two decades [1]. The recent Vizag gas leak killed at least 12 people and affected approximately 3000 others at the LG polymer chemical plant in Andhra Pradesh, India [2]. It is crucial to investigate not only the design and

operation of the process systems but also the safety protocols, such as emergency response and the evacuation of affected areas [3]. During an emergency evacuation, evacuees who receive detailed safety route information avoid hazards more than those who choose the preferred route [4]. However, most evacuation plans based on laws, codes, or government guidelines are too qualitative to be applied to unexpected practical situations [5]. To compensate for the limitations of traditional evacuation plans, several studies have focused on optimizing macroscopic evacuation, which deals with the movement of people as a flow, and developed algorithms to consider the structure of the accident zone [6], human behaviors [7], uncertainty [8, 9], evacuee characteristics [10], and operational costs [11]. While these macroscopic approaches deal with actual accident scenarios and the efficiency of the evacuation process itself, the output evacuation plans are not optimal for guaranteeing individual safety.

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<https://doi.org/10.1016/j.ress.2021.108102>

Received 19 January 2021; Received in revised form 17 September 2021; Accepted 24 September 2021

Available online 9 October 2021

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Nomenclature		$p_{\theta}(z \chi)$	intractable true posterior distribution in the variational autoencoder
Acronyms		$p_{\theta}(\chi z)$	likelihood of risk contour with latent variables in the decoder
ANOVA	analysis of variance	$q_{\phi}(z \chi)$	approximated posterior distribution in the variational autoencoder
CCD	central composite design	$r^{(i,j)}$	1 if edge (i, j) is selected for the evacuation route, otherwise 0
CNN	convolutional neural network	$SL^{(i,j)}$	segmented distance of edge (i, j) comparted by CFD space cells
DNN	deep neural network	$TD_t(x, y)$	toxic dose on the CFD space cell (x, y) at time t
GSA	global sensitivity analysis	$\overline{TD}_t(x, y)$	normalized toxic dose on the CFD space cell (x, y) at time t
MILP	mixed-integer linear programming	TD^{min}	maximum toxic dose in the whole data set
MSE	mean squared error	TD^{max}	minimum toxic dose in the whole data set
SGVB	stochastic gradient variational Bayes	$u_{t,h}^{(i,j)}$	1 if edge (i, j) and pre-combination h are selected at time domain t , otherwise 0
TDSPP	time-dependent shortest path problem	z	latent variables estimated in the encoder
VAE	variational autoencoder	Greek letters	
VAEDC	variational autoencoder with deep convolutional neural network	β	constants associated with the material in the Probit model for lethal toxicity
Sets		γ_t	time-dependent cost imposed on edge (i, j) associated with τ_j
$d \in D$	destination nodes (i.e., shelter candidates) in a graph network	$\tilde{\gamma}_t(x, y)$	concentration by volume within space coordinate (x, y)
$G(V, E)$	a graph composed of vertices V and edges E	$\epsilon^{(l)}$	normal distribution noise of sample l
$(i, j) \in E$	edges between node i and j	θ	generative parameters for true posterior in the variational autoencoder
$o \in O$	origin nodes (i.e., a node at starting evacuation) in a graph network	λ	surrounding information for the surrogate model (wind speed/direction, leak rate)
$\mathcal{P}$	set of nodes in the optimal evacuation path	λ^{data}	surrounding information for input data in training the surrogate model
$t \in T$	discrete time domains	λ^*	surrounding information for input variable in the surrogate model
$v \in V$	nodes (i.e., vertices) in a graph network	μ	mean in Gaussian distribution
$x \in X$	space coordinate of x-axis in CFD simulation cells	ρ	evacuation speed (i.e., evacuee's walking speed)
$y \in Y$	space coordinate of y-axis in CFD simulation cells	σ	standard deviation in Gaussian distribution
Roman letters		τ_v	elapsed time for traveling from the origin to node v
c_1, c_2, c_3	constants associated with the material in the Probit model for lethal toxicity	ϕ	variational parameters for approximated posterior in the variational autoencoder
$c(i, j, \tau_j)$	time-dependent cost imposed on edge (i, j) associated with τ_j	χ	contour matrices of normalized toxic dose within CFD cells
$C(x, y)$	concentration by volume within space coordinate (x, y)	χ_{gen}	contour matrices of normalized toxic dose generated in the decoder
$CE_{t,h}^{(i,j)}$	a pre-combined potential risk on the edge (i, j) by using snapshot potential risk	χ_{train}	contour matrices of normalized toxic dose calculated from the CFD simulation
D_{KL}	Kullback-Leibler divergence	$\psi^{(O,D)}$	potential risk from exposure to toxic material during traveling origin O to destination D
$f(\lambda)$	route optimization objective function with respect to surrounding information (λ)	$\tilde{\psi}^{(O,D)}$	approximated potential risk by discrete space and time domains
$I_t^{(O,D)}$	traveling distance between node O and D during time domain t	$\hat{\psi}_t^{(i,j)}$	approximated potential risk of time-domain t on edge (i, j) in the road network
$L^{(i,j)}$	distance on an edge (i, j) in the road network		
$\mathcal{L}(\theta, \phi; \chi)$	variational lower bound		
$m_v^{(i,j)}$	incidence parameter matrix representing a flow balance between edges		
n	index of training data of risk contour image in the variational autoencoder		
$\vec{p}(t)$	traveling curve of evacuation path on the field of risk function		
$p_{\theta}(z)$	prior distribution of latent variables in the variational autoencoder		

Conversely, quantitative risk measures are suitable indicators for identifying safe evacuation routes because they can assess alternate designs and are thus used in a relative risk approach [12]. Using numerical simulations of quantified risks, several studies have identified safe evacuation paths during fire incidents in offshore platforms [13], buildings [14], and underground mines [15]. In these studies, the indicator for quantitative risk depended on emergency needs. For example, visibility through smoke was used as a risk measure to optimize the evacuation path when conducting a fire simulation in a

building [14]. After adopting the appropriate risk indicator for a specific hazardous situation, risk-based evacuation planning was constructed following two modeling steps. First, a quantitative risk indicator of the affected area was evaluated, and an evacuation plan was then determined based on the results of risk evaluation [5]. Efforts have been exerted to combine the quantitative risk assessment of toxic gas release and the analysis of the affected area using the Gaussian dispersion model and societal risk indicator [16]. The Gaussian dispersion model was also applied to a multi-hazard analysis to identify zones with domino effects

using toxic-dose risk values [17].

Computational fluid dynamics (CFD) can evaluate toxic gas consequences as a risk indicator and offer the most accurate prediction of gas dispersion compared to other mathematical approaches, such as Gaussian plume and Lagrangian particle dispersion models, because it solves the Navier–Stokes equations for gas flow in complicated terrains [18]. Based on the outstanding accuracy of CFD modeling, numerous studies have attempted to utilize CFD results for toxic gas dispersion for various practical purposes in industrial facilities. Yang et al. evaluated the consequences of both toxicity and flammability on an offshore platform based on the output of a CFD dispersion simulation [19]. The mitigation effects of a physical barrier [20], a water curtain [21], and strategic decisions regarding the gas detector layout [22] can be assessed using three-dimensional CFD modeling of toxic and flammable gas dispersion. The CFD dispersion results can also be combined with the dose-response model in which the exposure time and toxic gas concentration determine the consequence of the absorbed dose to the human body. Yang et al. assessed human fatality using a Probit analysis for the Gumi hydrogen fluoride gas leak in South Korea by utilizing the CFD calculation output from an actual accident scenario [23]. The estimated CFD results aligned with the collected data of actual dead vegetation. Toxic load methods such as the acute exposure guideline level (AEGL) are also applicable when assessing human fatality using CFD results. Argyropoulos et al. analyzed fatality in terms of the Probit and AEGL methods from the CFD results for an evacuation scenario in which a building is infiltrated by toxic gas [24]. These toxic dose-response models can quantitatively evaluate the toxic gas consequences of pre-arranged evacuation routes by conducting several time steps of CFD simulation from when an individual starts evacuating to the end time [25].

Despite its high accuracy and applicability to risk evaluation, the CFD method still has shortcomings in assessing the quantitative risks of toxic gas dispersion incidents. Realistic atmospheric boundary layers, treatment of wind meandering effects, validation of simulation results, uncertainty from various sources, high costs, and long computational times are significant challenges limiting CFD simulation for investigating major hazards [26]. The issues noted above, except for computational costs and time, are related to the accuracy of the CFD model. A significant scholarly effort has been expended to improve the accuracy of each element noted above, for example, the specially tuned Reynolds-averaged Navier–Stokes (RANS) model for atmospheric boundary layers [27], modification of the dispersion model for reflecting wind meandering [26], utilization of experimental data for model validation [23, 28], and development of an accident sophisticated model for reducing uncertainty [29]. However, it is not only important to obtain accurate CFD data but also to improve its applicability, represented by its calculation time, to facilitate its use in real-time evacuation planning.

Quantitative risk studies that model complex physics have employed data-driven modeling to reduce high computational costs. According to a previous study on hazardous fire simulation, a machine learning data-driven model substantially reduced the computation time for estimating the significant conditions that start a fire from 1.5 d to a few seconds [30]. In atmospheric gas dispersion modeling based on CFD, this approach associates the input variables with the output variables represented as physical property contours or toxic consequences, which leads to building a risk prediction model without any prior knowledge [31]. One of the simplest data-driven models is an interpolation from the input variables of a leak scenario, such as the type of released material, leak position, leak rate, and weather conditions [32]. A least square vector machine and linear nearest neighbor interpolation were also utilized with CFD data for hazardous fire consequence analysis [33]. While surrogate modeling using simple regressions can predict the hazard distribution tendency, its accuracy needs to improve. In recent years, artificial neural networks, also known as deep neural networks (DNNs), have been studied to predict the consequences of hazardous gas

dispersion. Sun et al. developed a DNN to predict thermal radiation zones from fire scenarios involving several flammable chemicals [34]. Ni et al. used a deep belief network and convolutional neural network (CNN) to predict the concentration of released toxic gas [35]. Kim et al. applied a recurrent neural network with long short-term memory (LSTM-RNN) trained by CFD simulations to identify the location of hazardous material release [36]. Ma and Zhang combined a neural network with a Gaussian dispersion model to improve both computational efficiency and prediction accuracy when identifying an emission source [37]. Using dispersion simulation data, DNNs can predict gas concentration as well as other output variables, such as the maximum dispersion distance, plume size, and leaked gas arrival time at a specific position [38]. Na et al. proposed an autoencoder-based CNN to predict a lethality contour calculated from CFD and simulated a chlorine gas leak scenario in a petrochemical complex [39].

Although the development of a surrogate model for hazardous gas dispersion simulations has been broadly presented in previous studies, little research has been conducted on integrating these surrogate models to optimize evacuation route search. Lovreglio et al. evaluated the risk during evacuations using different time-domain CFD simulations, which consider individual human behaviors dynamically [40]. Wang et al. further considered different leak positions and weather conditions as input variables and assessed the evacuation response scenario [41]. However, these studies cannot offer specified safe evacuation routes for each evacuee considering individual situations because they used the A* algorithm-based evacuation software, which merely identifies the shortest path. Conversely, Liang et al. suggested an evacuation routing assignment scheme for a predefined road network and formulated it as a mixed-integer linear programming (MILP) problem [42]. However, a risk evaluation step was absent from their work. Cao et al. integrated evacuation planning and a cellular automata surrogate model to predict toxic gas dispersion [43]. However, their surrogate model cannot reflect the effects of terrain because only the convection–diffusion equation was considered without CFD simulation data. Gai et al. solved multi-objective optimization for determining evacuation routes under toxic gas dispersion to minimize both traveling time and distance using a modified Dijkstra algorithm [44]. However, only one leak scenario was considered and could not be applied in various situations because of the absence of a surrogate model. From an investigation of the literature related to the studies of hazardous gas dispersion modeling and evacuation route optimization, the challenges of establishing a practical and safety risk-based evacuation route search model can be summarized as follows:

- achieving acceptable prediction accuracy and moderate computation time;
- reflecting various variables in terms of the information specific to the incident region; and
- considering changes in gas dispersion over time.

In this study, we developed a framework for evacuation route optimization that meets the above requirements. To achieve high prediction accuracy and rapid computation, we utilized an autoencoder CNN-based surrogate model trained using CFD simulation data. Surrogate modeling should be elaborated as much as possible because the CFD risk contour data contain complex terrains and buildings that substantially affect the outcome of toxic gas dispersion. The autoencoder surrogate model can deal with elaborate CFD data without significant loss of detail, compared to other models such as principal component analysis (PCA), random forests, and regression kriging. Usually, it is beneficial to use PCA for linear datasets; the variance in the data is sufficiently explained without the loss of details [45]. However, the PCA method is difficult to use because of concerns about the loss of nonlinear CFD data during dimensionality reduction. Spatio-temporal regression kriging is available for surrogate modeling with CFD data; however, it is difficult to include the influences of surrounding information (i.e., wind speed,

direction, and leak rate) in the regression model. The autoencoder-based DNN model preserves data and extracts the important features that mainly influence the consequences of toxic gas leakage, such as the degree of dispersion, geography of the affected region, atmospheric conditions, and physical properties of the toxic material.

The dynamic behavior of the dispersed gas is reflected in the evacuation planning step by employing differently trained surrogate models for several time domains. This two-step framework, composed of predicting gas dispersion and proposing the safest route for individual evacuees, comprises an accelerated evacuation proposal system. We expect that this model will allow each evacuee to receive a specific evacuation route almost simultaneously with the alarm of toxic gas leakage. The novelty of the present study is that it attempts to use the variational autoencoder as a surrogate model for the safest route search in toxic gas leak scenarios. This study suggests time-dependent optimization to consider dynamic gas dispersion simultaneously during evacuation. Moreover, unlike previous research, we constructed an optimization framework with input variables such as wind speed, direction, leak intensity, and evacuee's location. Consequently, the proposed framework reduced the time taken for optimizing the evacuation route with an accuracy that is comparable to that of the CFD simulation. In addition, the results of this study leverage the model's short calculation time to enable further hazard analysis in the area affected by toxic gases. In Section 2 and 3, we describe the basic concept and structure of the proposed system and the risk indicator used. A sensitivity analysis of the input information surrounding the affected area was conducted, and the results are provided in Section 4.

2. Proposed accelerated evacuation route system

The evacuation system provides the safest route for evacuees from their position to the designated shelter. A risk-based evacuation route proposal system for the case of toxic gas dispersion follows two main procedures: (i) risk assessment of the target region and (ii) evacuation planning. Owing to the changes in gas dispersion under ambient conditions, a risk assessment should consider the dynamic behavior of the released toxic gas. It is also important to rapidly inform individuals of

the safety path before the beginning of the evacuation. The proposed system enables the prediction of a quantified risk contour posed by surrounding variables in the incident region within a few seconds and suggests the safest route via optimization. This system captures the surrounding information, such as weather conditions, type of hazardous materials, and incident scale, and converts it into the safest evacuation route for each evacuee's position. The conceptual diagram of the proposed system in Fig. 1 shows a sequence of information conversion from surrounding variables, GIS data, and extrinsic information to route optimization on the road network. CFD is used to simulate the quantitative risk contours from the dispersion of toxic gas because it can accurately predict the gas concentration at any point and time inside the computational domain [25]. The output data of a CFD sample is a quantitative risk indicator based on toxic concentration at the ground level, which takes a two-dimensional matrix perceived as a contour image. Therefore, the trained surrogate model naturally generates image-typed output data owing to the data format of the training dataset. The accuracy of the CFD simulation is adequate in terms of the numerical model as long as a safety engineer fully addresses uncertainties in the incident scenario [46]. However, the CFD simulation for a high-resolution contour of the gas concentration usually requires a long computational time, which prevents evacuees from making fast decisions. Hence, approximation approaches with affordable accuracy for both risk assessment and optimization procedures are required to guarantee real-time response and safety for all users.

2.1. Risk quantification

In the literature, the consequences of toxic exposures are mostly quantified by toxicologic criteria (e.g., ERPG, EEGL, or SPEGL) or the Probit model. The Probit model is appropriate for the measurement of lethality because the concentration of toxic material varies with time and weather conditions [47]. The probability of lethal toxicity is formulated in Eq. (1).

$$Probit = c_1 + c_2 \ln(TD) \quad (1)$$

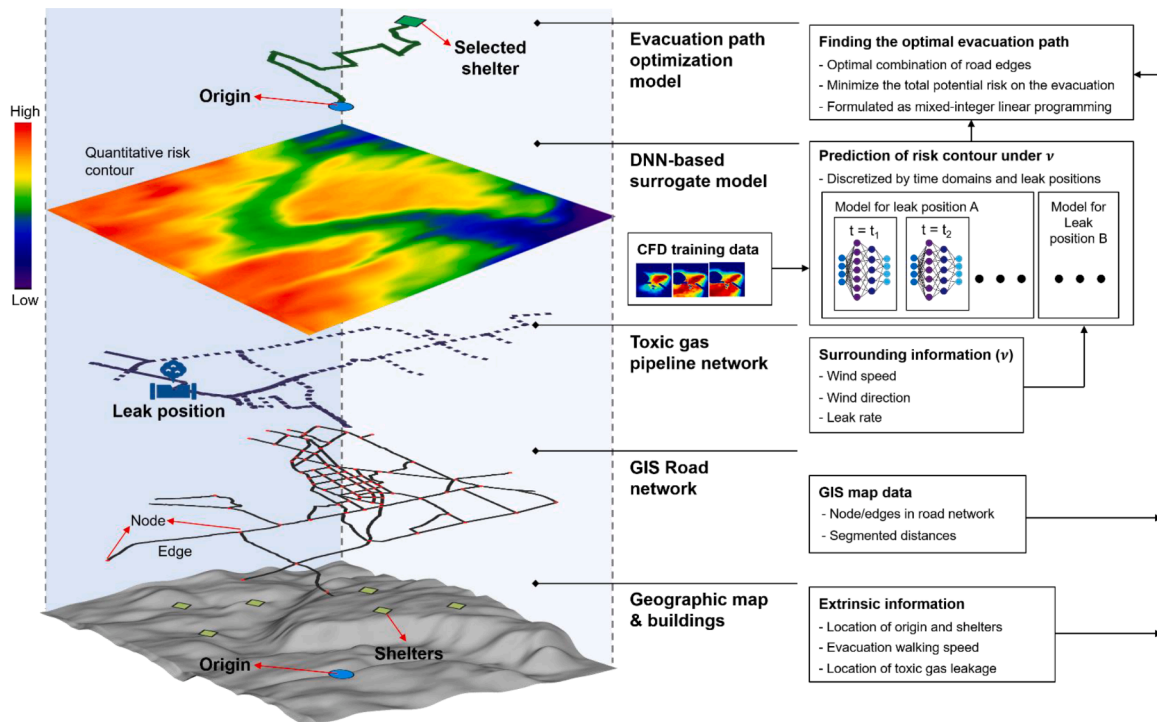


Fig. 1. Conceptual diagram of proposed accelerated evacuation route system for toxic gas leak incidents.

$$\tilde{\gamma}_t(x, y) = TD_t(x, y) = \int_{t_0}^t \{C(x, y)\}^{c_3} dt \quad (2)$$

Here, $\tilde{\gamma}_t(x, y)$ denotes a given quantitative risk value on a grid coordinate (x, y) at time t . The constants c_1 , c_2 , and c_3 are associated with the toxic material, and $C(x, y)$ is the concentration by volume within space coordinates (x, y) . We utilized the toxic dose (TD) defined by the product of the concentration of toxic gas and duration of exposure as the quantitative risk, which was also used in previous evacuation planning research [48]. The toxic dose $TD_t(x, y)$ is calculated by CFD simulation for space coordinates (x, y) and discrete time domain t , as formulated in Eq. (2). Then, it is normalized to the minimum and maximum range between 0 and 1 as follows:

$$\overline{TD}_t(x, y) = \frac{\log TD_t(x, y) - \log TD^{min}}{\log TD^{max} - \log TD^{min}} + \beta \quad (3)$$

$$TD^{min} = \min\{TD_t(x, y) | t \in T, x \in X, y \in Y, TD \neq 0\} \quad (4)$$

$$TD^{max} = \max\{TD_t(x, y) | t \in T, x \in X, y \in Y\} \quad (5)$$

where $\overline{TD}$ is the normalized toxic dose in the space coordinate (x, y) at time t , and TD^{min} and TD^{max} are the minimum and maximum toxic doses among the TD values calculated in the time and space domains. A small number β is added to the normalized TD to select a shorter distance path in the optimization if there is a small difference in the objective value between long- and short-distance paths. This normalization technique enhances the efficiency of the learning process of a DNN with nonlinear activation functions.

2.2. Real-time optimization strategy

The risk assessment was performed using a pretrained model, and it provided the predicted risk data to the evacuation planning algorithm. To achieve rapid risk assessment, a DNN was employed as a surrogate model to substitute the CFD simulation. We trained an independent DNN for different time domains to predict the risk contour that changed over time. The surrogate model calculates the quantitative risk values for different time domains within a space cell originally defined in CFD. MILP was used to identify the global optimum within an appropriate computation time limit for evacuation planning. It can aggregate the risk assessed from neural networks in different time domains. The evacuation planning method uses predefined road network information, a graph $G(V, E)$ that consists of nodes (V) and edges (E), to represent the evacuee's position and traveling path. Nodes and edges in a graph indicate intersections or shelters and roads, respectively, in a road network. The origin of an individual evacuee when the toxic gas is released is assumed to be a proximity node from the actual location. The optimal evacuation route is the combination of edges associated with the minimum risk to reach the shelter node from the origin node among every possible edge combination set.

We introduced the concept of potential risk to determine the evacuation path that has a relatively higher risk along the route from the origin to destination. Potential risk can be measured based on any type of quantitative risk, such as the toxic dose used in this study. If we assume that the evacuee's walking speed is constant during the evacuation, the potential risk accumulated along the evacuation route is proportional to the traveling distance and quantitative risk value calculated in the area corresponding to the location of the traveled path. The potential risk ψ from origin O to destination D to be minimized in the optimization model is defined as follows:

$$\psi^{(O,D)} = \int_{\tau_0}^{\tau_D} \gamma(\vec{p}(\tau)) \left\| \frac{d\vec{p}}{d\tau} \right\| d\tau \quad (6)$$

where γ is a scalar field of the distribution of quantitative risk, $\vec{p}(\tau)$ is the curve of the evacuation path at evacuation time τ since the occurrence of the incident, $\left\| \frac{d\vec{p}}{d\tau} \right\|$ is the Euclidean norm of the time derivative of the curve $\vec{p}$, and τ_0 , τ_D are the elapsed time at origin O and destination D during evacuation. The domain of γ is the area of risk assessment and can be divided by finite space cells for numerical computation, which is represented as a grid coordinate (x, y) . Time is also represented as a discrete domain; thus, the approximated form of Eq. (6) is given by:

$$\psi^{(O,D)} \cong \tilde{\psi}^{(O,D)} = \sum_{t \in T} \sum_{x \in X} \sum_{y \in Y} \tilde{\gamma}_t(x, y) \times l_t^{(O,D)}(x, y) \quad (7)$$

Here, $l_t^{(O,D)}(x, y)$ represents the length of the traveling path during time t within (x, y) . The surrogate model in the risk assessment process evaluates the quantitative risk value $\tilde{\gamma}_t(x, y)$ on the 2-dimensional cell coordinate (x, y) at each time stage t . Similarly, path curves on the risk field can be refined to the form of a road network that consists of nodes and edges, as illustrated in Fig. 1. Therefore, the MILP optimization process combines the edges to find a relatively smaller potential risk than that for all other possible combinations of edges, as formulated below:

$$\min_{\mathcal{P} \subseteq E} \sum_{(i,j) \in \mathcal{P}} \tilde{\psi}^{(i,j)} = \sum_{(i,j) \in \mathcal{P}} \sum_{t \in T} \sum_{x \in X} \sum_{y \in Y} \tilde{\gamma}_t(x, y) \times l_t^{(i,j)}(x, y) \quad (8)$$

where (i, j) is an edge on the road network expressed as node coordinates, $\mathcal{P}$ is an edge combination expressed as a minimum risk solution, and E is a set of all possible edge combinations to reach any shelter from the origin. The length of the path on the edge (i, j) within the grid coordinates (x, y) is denoted as $l_t^{(i,j)}(x, y)$. This approach discretizes the space and time domains and linearizes the calculation of the objective value (i.e., potential risk) in the optimization process. Consequently, the total computation time is efficiently reduced to indicate the optimal route for each evacuee for a real-time response.

3. Model framework

The computational model framework contains a description of two models and three types of input data. These two models are DNN-based surrogate models used for predicting the risk contour and MILP for determining an optimal path. Input data are categorized into extrinsic information, surrounding information, and GIS map data. Extrinsic information are the variables used in the entire framework to determine the boundary of the optimization problem. The location of the evacuee's origin, a set of candidate shelters, and evacuation walking speed define the problem statement in the optimization model. Similarly, the mode of the surrogate model is based on the location of toxic gas leakage; the model should be trained differently for different leak positions and time domains. In contrast, the surrounding information (λ) is the input variable used in the surrogate model. It depends on external factors surrounding the target region, such as weather conditions and accident scale. GIS map data are mainly represented as a road network, determined by the characteristics of the target region. The road network is a parameter set that includes nodes, edges, distances, and segmented distances. The following sections describe the structure of the models, the sampling method used for training the surrogate model, and the methods to input the three types of information into the models. Furthermore, we introduce the methods of sensitivity analysis for each input information on the total potential risk of the optimal evacuation route.

3.1. Neural network architecture

The risk assessment process in the proposed system utilizes a toxic dose contour, which can be depicted as a 2D image taken from the top

view in the zero z -direction. A variational autoencoder (VAE) with a deep CNN (VAEDC) can forecast contour images using different surrounding information (i.e., input variables, λ) [39]. Latent variables ($\mathbf{z}$) are extracted using VAE generated artificial data that resemble real data [49]. The developed surrogate model has a two-fold structure consisting of encoder and decoder parts, as shown in Fig. 2. The encoder part extracts the geometric features in the target region from the contour data χ , which is a matrix of the normalized toxic dose $\overline{TD}_t(x, y)$ in CFD cells. The variable layer, composed of a deep neural network between the encoder and decoder, maps the input variable space to the latent space. The decoder module generates the predicted contour data using latent variables calculated from the variational layer. In this scheme, the latent variable $\mathbf{z}$ is approximated by the observed data χ using a posterior distribution $p_\theta(\mathbf{z}|\chi)$, where θ is the generative parameter. Owing to the intractable true posterior $p_\theta(\mathbf{z}|\chi)$, its approximation $q_\phi(\mathbf{z}|\chi)$ with variational parameters ϕ is introduced in the encoder.

The parameters to estimate latent variables were optimized using the stochastic gradient variational Bayes (SGVB) method [49]. Maximizing the variational lower bound ($\mathcal{L}$) minimizes the loss function for training a VAE, which is expressed by the following equation:

$$\mathcal{L}(\theta, \phi; \chi^{(n)}) = -D_{KL}(q_\phi(\mathbf{z}|\chi^{(n)})||p_\theta(\mathbf{z})) + \mathbb{E}_{q_\phi(\mathbf{z}|\chi^{(n)})}[\log p_\theta(\chi^{(n)}|\mathbf{z})] \quad (9)$$

The term Kullback–Leibler divergence (D_{KL}) represents the regularization of the posterior approximation $q_\phi(\mathbf{z}|\chi)$ to the prior $p_\theta(\mathbf{z})$. The second term on the right-hand side of Eq. (9) indicates the reconstruction error between the n -th input contour data $\chi^{(n)}$ and the artificial data generated in the decoder $p_\theta(\chi^{(n)}|\mathbf{z})$. To differentiate the loss function for the learning process, a reparameterization trick is used to sample the latent variables $\mathbf{z}$ with a normal distribution noise (ϵ) using the

following equation:

$$\mathbf{z}^{(n,l)} = \boldsymbol{\mu}^{(n)} + \boldsymbol{\sigma}^{(n)} \odot \epsilon^{(l)}, \quad \epsilon^{(l)} \sim \mathcal{N}(0, \mathbf{I}) \quad (10)$$

where $\boldsymbol{\mu}^{(n)}$ and $\boldsymbol{\sigma}^{(n)}$ are the mean and standard deviation, respectively, estimated from the encoder. The latent variable $\mathbf{z}^{(n,l)}$ for data points n and l is generated using this method, which can transform the variational lower bound $\mathcal{L}(\theta, \phi; \chi^{(n)})$ to a tractable form for obtaining gradients. The mathematical formula of the SGVB estimator for the proposed neural network is detailed in the Supplementary Information.

The encoder part is composed of an input data layer, three convolutional layers with a pooling layer, two fully connected layers, a flatten layer, and mean ($\boldsymbol{\mu}$) and standard deviation ($\boldsymbol{\sigma}$) layers for sampling latent variables. The configuration of the decoder part was symmetrically reversed to the encoder. It has a latent variable ($\mathbf{z}$) layer linked from the encoder, two fully connected layers, a reshape layer, three deconvolutional layers, and a reconstruction layer that corresponds to the input data layer in the encoder. The hyperparameters and detailed configurations of each layer are summarized in Table S1. Four fully connected layers are arranged between the $\boldsymbol{\mu}$ layer of the encoder and the $\mathbf{z}$ layer in the decoder, which maps the input variables to the latent space. The input variable samples (λ^{data}) were used to train the fully connected layers with the generated contour data (χ_{gen}) from the decoder as the target output. These latent layers map the new input variable (λ^*) to the $\mathbf{z}$ layer in the decoder when we predict the unknown risk contour for the λ^* not included in the training dataset. Model training and inference were performed using the algorithm in Table 1.

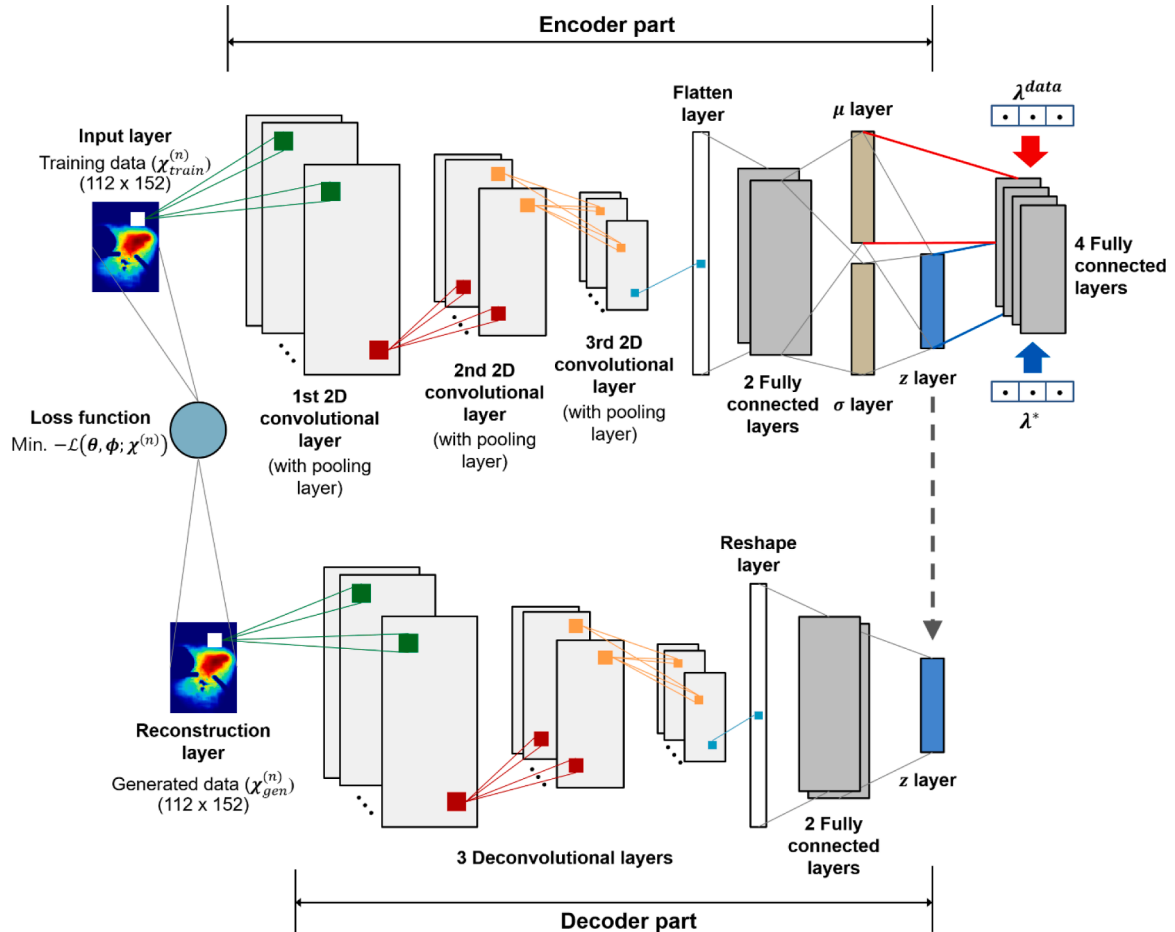


Fig. 2. Architecture of variational autoencoder with deep convolutional neural network for predicting toxic dose contour.

Table 1

Algorithm to train the VAEDC surrogate model.

	Data Generation & Training	Annotation
1:	for $LP = \{LP_1, \dots, LP_k\}$	- build different models for leak positions (LP)
2:	for $t = \{t_1, \dots, t_m\}$	- build different models time domains (t)
3:	initial sampling: LHS(dimension = 3, number of samples = 800) → λ^{data}	- latin hypercube sampling (LHS) for wind speed (m/s), wind direction (rad), and leak rate (kg/s)
4:	$Input(\lambda^{data}) \rightarrow$ FLACS simulation $\rightarrow Output(\chi^{data})$	- toxic dose ($\overline{TD}_t(x, y)$) calculation
5:	$\mathcal{D}(\lambda^{data}, \chi^{data}) \rightarrow$ Data split $\rightarrow \{\mathcal{D}_{train}, \mathcal{D}_{validation}, \mathcal{D}_{test}\}$	- split ratio: = {600, 150, 50}
6:	encoding: $Input(\chi^{data, train}) \rightarrow$ Input layer $\rightarrow$ 3 Convolutional layers $\rightarrow$ Flatten layer $\rightarrow$ 2 Fully connected layers $\rightarrow Output(\mu, \sigma)$	- extract μ, σ from the $\chi^{data, train}$
7:	reparameterization: random normal sampling $\varepsilon^{(l)} \sim \mathcal{N}(0, I)$	
8:	$z^{(n,l)} := \mu^{(n)} + \sigma^{(n)} \odot \varepsilon^{(l)}$	
9:	decoding: $Input(z) \rightarrow$ 2 Fully connected layers $\rightarrow$ Reshape layer $\rightarrow$ 3 Deconvolutional layers $\rightarrow$ Reconstruction layer $\rightarrow Output(\chi^{gen})$	- generate $\chi^{gen, train}$ from the μ, σ
10:	KL loss:= $-\sum_{n=1}^{N_{train}} \sum_{i=1}^{N_t} \frac{1}{2N_{train}} (1 + \log(\sigma_i^{(n)})^2 - (\mu_i^{(n)})^2 - (\sigma_i^{(n)})^2)$	(supplementary information)
11:	reconstruction loss:= $-\frac{1}{N_{train}} \sum_{n=1}^{N_{train}} \sum_{x=1}^{152} \sum_{y=1}^{112} [\chi_{(n,x,y)}^{data, train} \log \chi_{(n,x,y)}^{gen} + (1 - \chi_{(n,x,y)}^{data, train}) \log(1 - \chi_{(n,x,y)}^{gen})]$	(supplementary information)
12:	VAE training: minimize (KL loss + reconstruction loss)	- Adam optimizer (learning rate=0.002)
13:	mapping layer training: $Input(\lambda^{data}) \rightarrow$ 4 Fully connected layers $\rightarrow Output(\mu^{mapping})$	
14:	minimize $\sum_{n=1}^{N_{train}} \sum_{i=1}^{N_t} \frac{1}{N_{train}} (\mu_i^{(n), encoder} - \mu_i^{(n), mapping})^2$	- Adam optimizer (learning rate=0.002)
15:	next t	
16:	next LP	
	Inference	- generate χ^{gen} from the given λ^* using the trained VAEDC
17:	input:= (LP, t, λ^*)	
	mapping surround information:	
18:	$Input(\lambda^*) \rightarrow$ 4 Fully connected layers $\rightarrow Output(\mu)$	
	reparameterization:	
19:	random normal sampling $\varepsilon^{(l)} \sim \mathcal{N}(0, I)$	
20:	$z^{(n,l)} := \mu^{(n)} + \sigma^{(n)} \odot \varepsilon^{(l)}$	
	decoding:	
21:	$Input(z) \rightarrow$ 2 Fully connected layers $\rightarrow$ Reshape layer $\rightarrow$ 3 Deconvolutional layers $\rightarrow$ Reconstruction layer $\rightarrow Output(\chi^{gen})$	
22:	output:= $\chi^{gen} \in \mathbb{R}^{112 \times 152}$	- return predicted toxic dose ($\chi^{gen} = \overline{TD}_t(x, y)$) at time t

3.2. Optimization model

The optimization model is based on a time-dependent shortest path problem (TDSP) [50]. Given graph $G(V, E)$ with cost function $c(i, j, \tau_j)$ imposed on E , nodes $v \in V$, edges $(i, j) \in E$, an origin node $o \in O$, a destination node $d \in D$, and elapsed time (τ_v) at node v , TDSP finds a path $\mathcal{P} = \{i_1(=o), \dots, i_k(=d)\}$ to minimize the time-dependent cost.

$$f(\lambda) = \min \sum_{(i,j) \in \mathcal{P}} c(i, j, \tau_j) \times u^{(i,j)} \quad (11)$$

$$subject\ to \sum_{(i,j) \in E} m_v^{(i,j)} \times u^{(i,j)} \forall v \in V \quad (12)$$

$$\left(\tau_i + \frac{L^{(i,j)}}{\rho} \right) \times u^{(i,j)} \leq \tau_j \forall (i,j) \in E \quad (13)$$

$$u^{(i,j)} \in \{0, 1\} \forall (i,j) \in E \quad (14)$$

$$\tau_v \geq 0 \forall v \in V \quad (15)$$

The objective value f is minimized using the given input variable λ . The incidence parameter of the given graph, $m_v^{(i,j)} \in \{-1, 0, 1\}$, represents the flow balance between edges that takes the value -1 if $v=j$, $+1$ if $v=i$, and 0 otherwise. The elapsed time (τ_v) is determined by the distance $L^{(i,j)}$, the evacuee's walking speed (ρ) is assumed to be constant, and the selected path prior reaches node v in the above formulation. However, we applied a piecewise linearization approach to the cost function $c(i, j, \tau_j)$ because the calculation of the elapsed time at each node is nonlinear in integer programming, and the risk assessment is carried out in discrete time domains. This approach obtains the fixed-time costs on each edge in different time domains and combines these costs into several piecewise cases classified by walking speed and elapsed time [51]. Then, the integer programming selects an appropriate combined cost to be imposed for each edge through the constraints of the time sequence at each node.

The surrogate model conveys the predicted quantitative risk matrix ($\overline{TD}_t(x, y)$) to the optimization model, which identifies the optimal combination of road edges using binary variables in the branch-and-cut method [52]. Substituting a continuous decision variable $l_t^{(i,j)}(x, y)$ in Eq. (8) to the integer type, the snapshot potential risk (i.e., fixed-time cost) is introduced. The segment distance ($SL^{(i,j)}(x, y)$), which indicates the partial length of $L^{(i,j)}$ within the CFD cell coordinate (x, y) , is used to calculate the snapshot potential risk ($\hat{\psi}_t^{(i,j)}$) on the edge (i, j) assessed at time t .

$$\hat{\psi}_t^{(i,j)} = \sum_{x \in X} \sum_{y \in Y} SL^{(i,j)}(x, y) \times \overline{TD}_t(x, y) \quad (16)$$

This type of potential risk reflects the toxic dose obtained in time domain t . The actual potential risk ($\tilde{\psi}^{(i,j)}$) that reflects the change in time along the edge (i, j) can be approximated by combining the snapshot potential risks of different time domains. We assumed that a time-dependent cost function $c(i, j, \tau_j)$ is piecewise constant in the time domains and reformulated as follows [53]:

$$c(i, j, \tau_j) = \begin{cases} c_1(i, j, \tau_j), & 0 \leq \tau_j \leq t_1 \\ c_2(i, j, \tau_j), & t_1 \leq \tau_j \leq t_2 \\ c_3(i, j, \tau_j), & t_2 \leq \tau_j \leq t_3 \\ \dots & \dots \end{cases} \quad (17)$$

The elapsed time of evacuation (τ_j) at destination node j depends on the elapsed time at a prior node (τ_i), distance $L^{(i,j)}$, and evacuation speed (ρ). The basis for piecewise linearization is that all possible potential risks are defined on the edge (i, j) , and constraints are screened out. The predefining method is a combination of snapshot potential risks $\hat{\psi}_t^{(i,j)}$ in different time domains based on the criteria of τ_i and $L^{(i,j)}/\rho$. As a result, the pre-combined potential risk, $CE_{t,h}^{(i,j)}$ is introduced into the integer programming. Thus, the objective function with binary variables for integer programming is given as follows:

$$\min \sum_{(i,j) \in \mathcal{P}} \tilde{\psi}^{(i,j)} \cong \sum_{(i,j) \in \mathcal{P}} \sum_{t \in T} \sum_{h \in H} CE_{t,h}^{(i,j)} \times u_{t,h}^{(i,j)} \quad (18)$$

where $CE_{t,h}^{(i,j)}$ is the pre-combined potential risk on the edge (i, j) using

snapshot potential risk, and $u_{t,h}^{(ij)}$ is a binary decision variable that represents the selection of edge (i, j) . The subscripts t and h determine the combination of the snapshot potential risk values, which depends on the evacuee's walking speed and the elapsed time at each node.

Given a graph $G(V, E)$, four time domains $\{t_1, t_2, t_3, t_4\}$, snapshot potential risks $\hat{\psi}_t^{(ij)}$, edge distance $L^{(ij)}$, evacuation speed ρ , and interval of time domains Δt , the pre-combined potential risk $CE_{t,h}^{(ij)}$ on edge (i, j) is defined by the following equations:

$$CE_{t,h_1}^{(ij)} = \hat{\psi}_t^{(ij)}, \forall t = t_1, t_2, t_3, t_4 \quad (19)$$

$$CE_{t,h_2}^{(ij)} = \hat{\psi}_t^{(ij)} \frac{\rho \Delta t}{L^{(ij)}} + \hat{\psi}_{t+1}^{(ij)} \frac{L^{(ij)} - \rho \Delta t}{L^{(ij)}}, \forall t = t_1, t_2, t_3 \quad (20)$$

$$CE_{t,h_3}^{(ij)} = \left(\hat{\psi}_t^{(ij)} + \hat{\psi}_{t+1}^{(ij)} \right) \frac{\rho \Delta t}{L^{(ij)}} + \hat{\psi}_{t+2}^{(ij)} \frac{L^{(ij)} - 2\rho \Delta t}{L^{(ij)}}, \forall t = t_1, t_2 \quad (21)$$

$$CE_{t,h_4}^{(ij)} = \left(\hat{\psi}_t^{(ij)} + \hat{\psi}_{t+1}^{(ij)} + \hat{\psi}_{t+2}^{(ij)} \right) \frac{\rho \Delta t}{L^{(ij)}} + \hat{\psi}_{t+3}^{(ij)} \frac{L^{(ij)} - 3\rho \Delta t}{L^{(ij)}}, \forall t = t_1 \quad (22)$$

The above equation for the index $t = t_a$ indicates that the elapsed time at a prior node (τ_i) ranges from t_{a-1} to t_a , i.e., $t_{a-1} \leq \tau_i < t_a$. The indices h_1 – h_4 represent an internal point of division to set weights of snapshot potential risks corresponding to the elapsed time during the passage from node i to j . In other words, long-distance or slower walking speeds should linearly combine the snapshot potential risk for more time domains.

The integer programming with the pre-combined potential risk $CE_{t,h}^{(ij)}$ introduces three types of binary variables, $u_{t,h}^{(ij)}$, $r^{(ij)}$, w_i , and one continuous variable, τ_i . The binary variable $u_{t,h}^{(ij)}$ determines the cost type that is added to the objective function under a given τ_i and $L^{(ij)}$. Meanwhile, binary variable $r^{(ij)}$ is the actual solution to determine whether the edge (i, j) is included in the optimal path $\mathcal{P}$, regardless of the time domain. Only one destination node is selected as the optimal shelter among candidates $d \in D \subset V$ using a binary variable w_i . The elapsed time at node i (τ_i) is calculated using the constraint. Therefore, the TDSPP presented in this study is piecewise linearized and reformulated using the following equations:

$$\min \sum_{(i,j) \in \mathcal{P}} \sum_{t \in T} \sum_{h \in H} CE_{t,h}^{(ij)} \times u_{t,h}^{(ij)} \quad (23)$$

Subject to

$$\sum_{(i,j) \in E} r^{(ij)} - r^{(j,i)} = 1, \forall i \in O \quad (24)$$

$$\sum_{(i,j) \in E} r^{(ij)} - r^{(j,i)} = 0, \forall i \in V \setminus (O \cup D) \quad (25)$$

$$\sum_{(i,j) \in E} r^{(ij)} - r^{(j,i)} = -w_i, \forall i \in D \quad (26)$$

$$\sum_{i \in D} w_i = 1 \quad (27)$$

The above constraints meet a flow balance and selection of only one optimal shelter.

$$\tau_i + \frac{L^{(ij)}}{\rho} = \tau_j, \forall (i,j) \in E : r^{(ij)} = 1 \quad (28)$$

$$\tau_i = 0, \forall i \in O \quad (29)$$

$$\tau_i \leq \tau_i^{\max}, \forall i \in V \quad (30)$$

The above constraints are used to calculate the elapsed time at each node in the optimal path $\mathcal{P}$. A conditional statement can be transformed

into inequality constraints by adding a new binary variable and an arbitrarily large constant [54]. To match the activation of $u_{t,h}^{(ij)}$ and $r^{(ij)}$, the following equations are added to the constraints using a conditional statement.

$$u_{t_1,h}^{(ij)} = r^{(ij)}, \forall (i,j) \in E : \tau_i \leq \Delta t \quad (31)$$

$$u_{t_2,h}^{(ij)} = r^{(ij)}, \forall (i,j) \in E : \Delta t + \varepsilon \leq \tau_i \leq 2\Delta t \quad (32)$$

$$u_{t_3,h}^{(ij)} = r^{(ij)}, \forall (i,j) \in E : 2\Delta t + \varepsilon \leq \tau_i \leq 3\Delta t \quad (33)$$

$$u_{t_4,h}^{(ij)} = r^{(ij)}, \forall (i,j) \in E : 3\Delta t + \varepsilon \leq \tau_i \quad (34)$$

These constraints determine the $CE_{t,h}^{(ij)}$ that is appropriate for time-dependent cost, based on the elapsed time at node i . An arbitrary small number ε is used to avoid disjunctive inequality ($<$). Conversely, the subscript index h is determined by the following equation:

$$\sum_{t \in T} u_{t,h_1}^{(ij)} = r^{(ij)}, \forall (i,j) \in E : \frac{L^{(ij)}}{\rho} \leq \Delta t \ \& \ \tau_i \leq 4\Delta t \quad (35)$$

$$\sum_{t \in T} u_{t,h_2}^{(ij)} = r^{(ij)}, \forall (i,j) \in E : \Delta t + \varepsilon \leq \frac{L^{(ij)}}{\rho} \leq 2\Delta t \ \& \ \tau_i \leq 4\Delta t \quad (36)$$

$$\sum_{t \in T} u_{t,h_3}^{(ij)} = r^{(ij)}, \forall (i,j) \in E : 2\Delta t + \varepsilon \leq \frac{L^{(ij)}}{\rho} \leq 3\Delta t \ \& \ \tau_i \leq 4\Delta t \quad (37)$$

$$\sum_{t \in T} u_{t,h_4}^{(ij)} = r^{(ij)}, \forall (i,j) \in E : 3\Delta t + \varepsilon \leq \frac{L^{(ij)}}{\rho} \leq 4\Delta t \ \& \ \tau_i \leq 4\Delta t \quad (38)$$

To reduce the size of the search space, the contradicting activation of $u_{t,h}^{(ij)}$ is set to zero in the following equations:

$$\sum_{t \in T} u_{t,h}^{(ij)} \leq r^{(ij)}, \forall (i,j) \in E \quad (39)$$

$$u_{t_4,h_2}^{(ij)} = 0, \forall (i,j) \in E : \tau_i \leq 4\Delta t \quad (40)$$

$$u_{t_3,h_3}^{(ij)} = 0, \forall (i,j) \in E, t \in \{t_3, t_4\} : \tau_i \leq 4\Delta t \quad (41)$$

$$u_{t_4,h_4}^{(ij)} = 0, \forall (i,j) \in E, t \in \{t_2, t_3, t_4\} : \tau_i \leq 4\Delta t \quad (42)$$

$$u_{t,h}^{(ij)} = 0, \forall (i,j) \in E, t \in \{t_1, t_2, t_3\}, h \in \{h_2, h_3, h_4\} : 4\Delta t + \varepsilon \leq \tau_i \quad (43)$$

4. Case study

4.1. Study area and sampling

The optimization model of the evacuation route was developed and tested for the Mipo industrial complex located in Ulsan, South Korea. The scale of the CFD simulation was 2800, 3800, and 80 m in the x , y , and z directions, respectively. Toxic dose values in a space cell with a uniform grid resolution of 25 m in the x and y directions were obtained from the CFD simulation at the ground level. The three variables, wind speed (1.520 m/s), wind direction (0 – 2π), and leak rate (50–150 kg/s), were selected as the surrounding information because an evacuee's ability to escape mainly depends on the size of the toxic gas cloud and wind speed [47]. The leaked toxic material was assumed to be ammonia gas in the incident scenario, and four specific leak positions close to the residential area were chosen to examine the proposed system. The toxic dose for each space cell (x, y) was measured at four different time domains at 5 min intervals. Flame acceleration simulator (FLACS), which is a specialized CFD simulator for gas dispersion, explosion, and fires, was used to generate a training dataset (i.e., normalized toxic dose contours) to construct the surrogate model. It includes sophisticated analytical and

empirical models including mass and momentum conservation, the $k-\epsilon$ turbulence model, wall functions, and wind boundaries. The wind boundary equations calculate the buoyancy effects on the atmospheric boundary layer in terms of the velocity profile in the z direction. The output data from the CFD simulation contained latent features that were extracted by the surrogate model. The governing equations of the CFD model are provided in the supplementary information. The ranges of the variables and scenario conditions for the FLACS simulation are listed in Table 2.

For each time domain and leak position, the number of datasets used for the training, validation, and testing of each deep neural network were 600, 150, and 50, respectively. The Latin hypercube sampling method was employed to generate 800 input variable samples (i.e., surrounding information λ , wind speed, wind direction, and leak rate) used in the FLACS simulation for each time domain. The total number of samples for training and testing was $800 \times 4 \times 4$ for four time domains and four leak positions. The format of the output sample data is a 112×152 matrix in which each element represents the toxic dose measured within space coordinates (x, y) corresponding to the matrix's row and column number.

4.2. Road network

We obtained the geographical information of the road network in the target region of the Mipo industrial complex and combined it to the location and arrangement of chemical and gas pipelines, as shown in Fig. 3(a). For the graph configuration, the road network was divided into 85 nodes and 134 edges. The distance of every edge (i, j) segmented by space cells (x, y) was measured and converted to a matrix form ($SL^{(ij)} \in \mathbb{R}^{112 \times 152}$) to compute the potential risk.

4.3. Sensitivity analysis

The optimization model suggests the safest route considering the given decisive variables within a few seconds. This enables a comprehensive parametric study that requires simulation data to determine how variables influence evacuation risks in the target region. To run the optimization model, three types of decisive variables are needed: surrounding information used in the surrogate model (λ : wind speed, wind direction, and leak rate), the leak position that constructs independent surrogate models for each position, and origin and shelter locations represented as nodes in the road network. To identify the variable that significantly affects the potential risk on the optimal evacuation route, we employed a variance-based global sensitivity analysis method for continuous decisive variables (λ) and set up the design of experiments for discrete types (leak positions, origins, and shelters).

We examined the quantified contribution of each input variable (λ) to the total variance of the optimal objective value, $a = f(\lambda)$, in Eq. (11). The other input parameters (locations of origin, shelters, and leak positions) were fixed in a specific categorized group. The variance-based

first-order sensitivity index S_j is given by the ratio of the conditional variance to the total variance of the objective function [55]. The index represents the contribution of the input variable to the output function f , excluding any interactions with other variables. The mathematical formula for the global sensitivity analysis (GSA) is provided in detail in the Supplementary Information. We repeated the GSA with 301 samples of the input variables (ν) for the location of the origin node number {5, 12, 16, 31, 53, 57} and the location of the leak position {13, 15}. For each case, the candidate shelters were chosen to be evenly located in the target region. The locations for the origin, shelter, and leak position used in the GSA are shown in Fig. 4.

To extend the sensitivity analysis for all origin and shelter candidate sets, we roughly designed sample points using the central composite design (CCD) method. This approach reduces the number of running optimization models. The CCD is a two-level factorial design with the addition of $2^n + 2n + 1$ experiments with respect to n independent variables. A three-dimensional inscribed type CCD is applied to the range of wind speed, wind direction, and leak rate; hence, 15 samples are required for each origin node and each leak position. All nodes in the road network were considered as shelter candidates. For 85 origin nodes and 4 leak positions, 5100 cases were optimized. The sampling results of the input variables and the categorized cases are summarized in Table 3. We examined the potential risk of each shelter candidate position by taking an average of all cases.

5. Results and discussion

5.1. Model performance for searching optimal path

The accuracy of the surrogate model was measured by determining the mean squared error (MSE) of the test set data. The accuracy of the other surrogate models including PCA and CNN-based autoencoder (CNN-AE) were compared to the VAEDC (CNN-VAE). The number of latent variables (i.e. reduced dimensions) is set to 16, 32, and 64 in the three models. Those numbers have the explained variance ratio higher than 99% in the PCA transform of the training data set. The PCA model reduces the dimension of normalized toxic dose contours (χ). The CNN-AE model has the same architecture as the CNN-VAE except for the removal of the variational layers (μ, σ) and replacement of the loss function ($-\mathcal{L}$) into MSE. For each model, four fully connected layers were trained to predict the latent variables from the input variables of surrounding information (λ). The distribution of MSE values of the prediction of test samples for node #10 leak position and 5 min time domain is shown in Fig S1. Autoencoders are analogous to a PCA if a linear activation function is used. Since the autoencoder-based models adopted nonlinear ReLU activation functions and dealt with the nonlinearity of the data, they have lower MSE values than PCA models. Also, the variational inference in VAE models seems to allow reducing the MSE variance of the test samples. While the MSE values and standard deviation are decreased in PCA and CNN-AE models as the number of latent variables increases, the CNN-VAE with 64 latent variables rather decreases model accuracy compared to the models with 16 and 32 latent variables. Therefore, the CNN-VAE with 32 latent variables is the most suitable model in terms of a lower mean MSE (0.0043) and a smaller standard deviation (0.0037) of the test samples.

The MSE values for each time domain and leak position are shown in Fig. S2. Since the evacuation proposal system selects a solution that has a relatively lower potential risk, based on the normalized toxic dose, than the other solutions, a mean MSE value of less than 0.02 appears to be an acceptable prediction error to be used in the optimization model. The risk contour images of some test samples for each leak position are shown in Fig. S3. The difference contours between the predicted and actual values in Figs. S3(c), (f), (i), and (l) are negligible for calculating the potential risk cumulated on evacuation paths.

The computation time from the prediction of the potential risk to the optimization was less than 10 s. This duration is the time required to

Table 2
Variable ranges and scenario conditions for FLACS simulation.

Item	Attribute	Unit	Value
Simulation settings	Map scale	m	2800 × 3800 × 80
	Grid resolution	m	25
	Toxic specification	–	Ammonia
	Leak duration	s	120
	Monitoring time	min	5 / 10 / 15 / 20
	Number of leak positions	–	4
Ambient conditions	Pressure	bar	1
	T emperature	°C	20
	Pasquill class	–	Neutral
Surrounding information	Wind speed	m/s	1.5–20
	Wind direction	radian	0–2 π
	Leak rate	kg/s	50–150

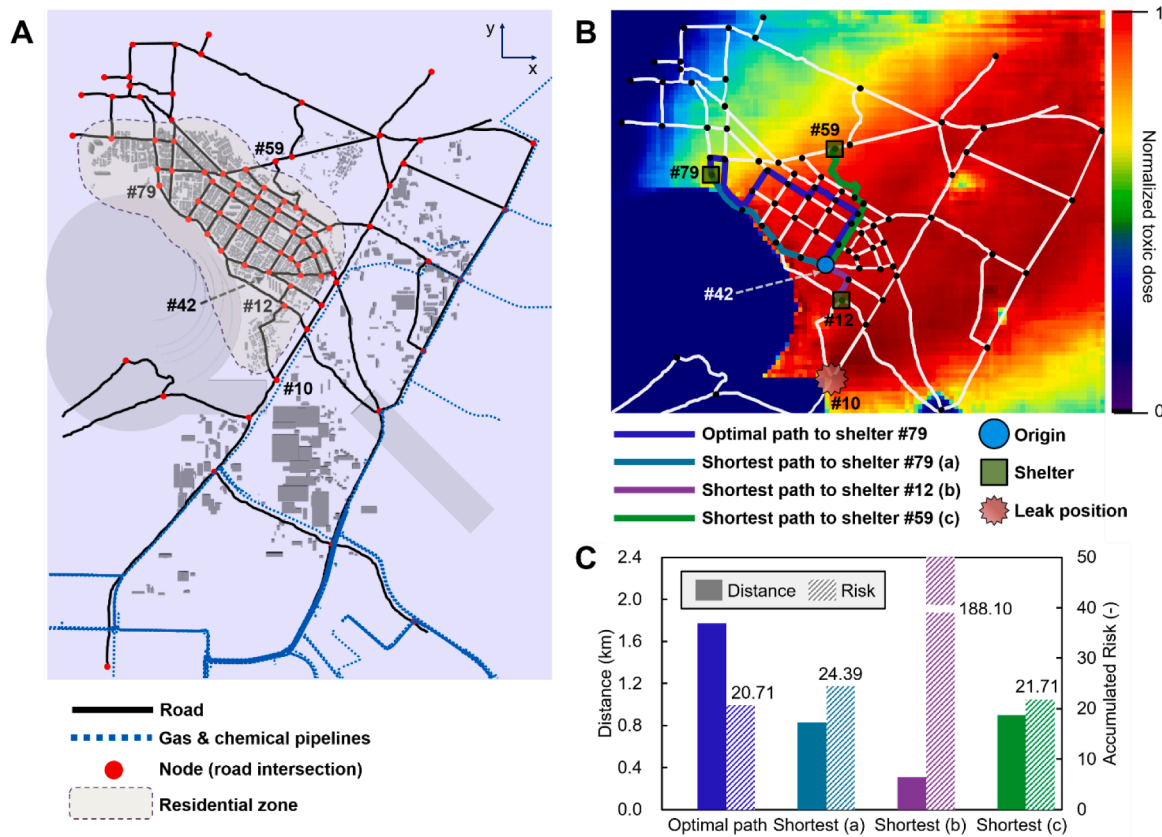


Fig. 3. (a) Road network, pipelines, and residential zone of the Mipo industrial complex region. (b) Case study results for optimal evacuation route determination under the following conditions: evacuation walking speed of 6 km/h, wind speed of 6.5 m/s, wind direction of 13.8°, and ammonia leak rate of 88.9 kg/s. (c) Comparison of total traveling distance and accumulated potential risk on the optimal path: shortest paths a, b, and c.

calculate the optimal evacuation route on a computer. Leak detection, sensing input variables, and alarm processes were not considered in this study. However, we can reflect the time delay due to the external safety systems if the system knows the incident time, by adjusting different time domains in the prediction risk contours of the surrogate model. For example, if there is a delay of 10 min owing to additional safety systems, the model replaces the 15 min prediction result with a risk contour at 5 min. Moreover, the large area of the petrochemical complex implies that most evacuees have sufficient time to receive optimal evacuation information in 10 s. According to the simulation results of toxic gas leak incidents in a petrochemical complex, the time taken for toxic gas to reach each affected building is usually over a minute [56].

The developed model was tested for a specific incident scenario in which the evacuation speed, wind speed, wind direction, and leak rate were fixed as 1.7 m/s, 6.5 m/s, 13.8 deg, and 88.9 kg/s, and the leak position was node 10, as shown in Fig. 3(a). Nodes #12, #59, and #79 were assumed to be candidate shelters. The optimal evacuation path from origin #42 to the shelter is marked in blue in Fig. 3(b), which is a distance of 1.77 km to shelter #79. The distance of the shortest path, marked in blue-green, from the origin to shelter #79 is 0.82 km, which is shorter than the optimal path; however, the potential risk is 3.68 higher than the optimal path, which has a potential risk of 20.71. The other shortest routes to shelters #59 and #12 also take shorter distances than the optimal path, as shown in Fig. 3(c). Although shelter #12 is the closest to the origin, the optimization model proposes a path to shelter #79 with a longer distance because the route to shelter #12 exhibits a potential risk that is 9.1 times higher than the optimal path. This implies that the optimal evacuation route provided by the accelerated alarm system ensures the safest path rather than the user's intuitive decision to move to the closest shelter in the case of toxic gas leakage.

To compare the results obtained using changing variables, we

designed six scenarios with different input information, as shown in Table 4. The candidate shelters were nodes #59, #74, and #79. The proposed model suggests different optimal paths according to the input information, as shown in Fig. 5(a). The snapshots for each case in Fig. 5 (b) are the predicted risk contours at the time when an evacuee arrives at the shelter. A change in the evacuee's walking speed in Case 1 maintains the same predicted risk contour as the base case; however, a slow evacuation results in the selection of a closer shelter than in the base case. A change in the input variables in the surrogate model affects the configuration of the risk contours. Changing the wind speed (Case 2) and the leak rate (Case 4) led to slightly different risk contours from the base case; however, the optimization model provided completely different paths and shelters. The optimization model selects the shorter distance to shelter (#59) in Case 2 with a 6.03 higher potential risk than the base case, which reduces exposure time against toxic gas spread by a strong wind. In contrast, the optimal path of Case 4 detours around the shelter because a higher leak rate increases the potential risk in the overall region, while the leaked gas reaches a narrow area due to low wind speed. Changing the wind direction (Case 3) and leak position (Case 5) formed different risk contours compared with the base case. The optimal paths in these two cases select the closest shelter from the origin to avoid a long exposure in the high-risk area.

The evacuation paths in Figs. 3 and 5 show that the proposed optimization model offers the safest route with the lowest potential risk; however, it is contrary to the intuitive bias, such as the shortest path to the shelter. In addition, the surrogate model enables individuals to respond to diverse incident scenarios within a few seconds of incident occurrence.

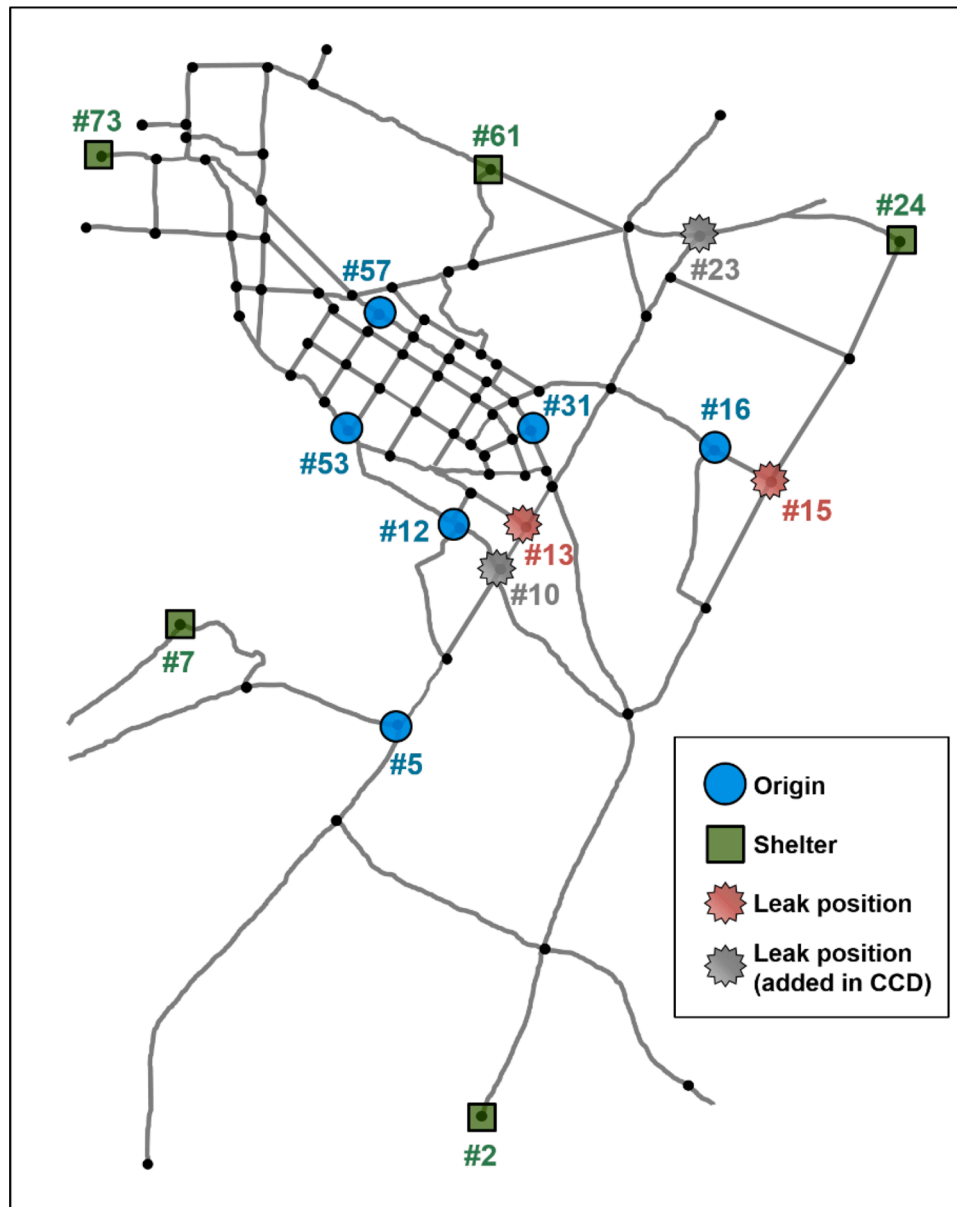


Fig. 4. Designated evacuee's origins, candidate shelters, and leak positions for global sensitivity analysis (GSA) and central composite design (CCD) studies: origins #5, #12, #16, #31, #53, #57, shelters #2, #7, #24, #61, #73, and leak positions #13, #15 for GSA; origins #1–#85, shelters #1–#85, leak positions #10, #13, #15, #23 for CCD.

Table 3

Sampling result of central composite design for determining optimal shelter.

Variable type		Samples
Continuous		(wind speed (m/s), wind direction (rad), leak rate (kg/s))
		{(5.25, 1.27, 70.27), (5.25, 1.27, 129.73), (5.25, 5.01, 70.27), (5.25, 5.01, 129.73), (16.25, 1.27, 70.27), (16.25, 1.27, 129.73), (16.25, 5.01, 70.27), (16.25, 5.01, 129.73), (1.50, 3.14, 100), (20, 3.14, 100), (10.75, 0, 100), (10.75, 6.28, 100), (10.75, 3.14, 50), (10.75, 3.14, 150), (10.75, 3.14, 100)}
Categorical	Leak position	{#10, #13, #15, #23}
	Origin	{#1, #2, #3, ..., #85}

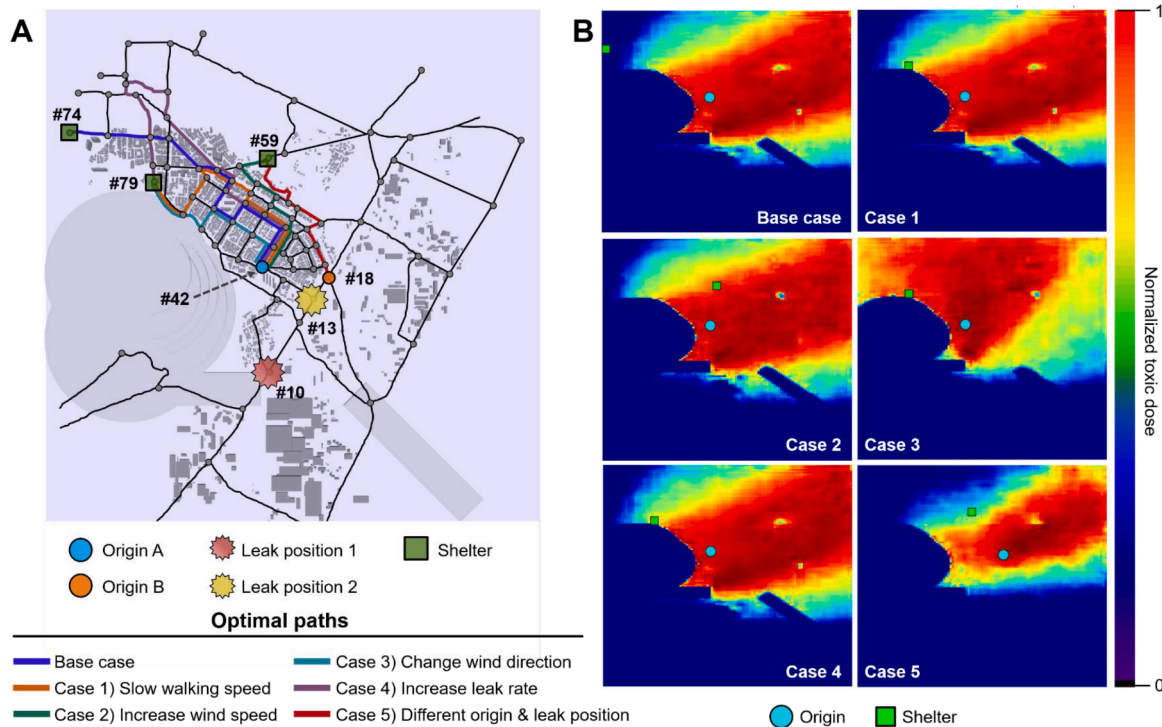
5.2. Influences of variables

In the proposed framework, the DNN-based surrogate model extracts the geometric features as a risk contour under the given surrounding information ν . The road network represented as a graph $G(V, E)$ reflects the connectivity of roads and population density implied by the compactness of the node arrangement. The features extracted from the surrogate model and road network inherently influence the potential risk of evacuation routes. In addition to the influence of these inherent features, the potential risk value on an evacuation route varies with the given input information such as wind speed, wind direction, leak rate, location of origin, candidate shelters, and leak position. To investigate the influences of the input variables, sensitivity indices for wind speed, wind direction, and leak rate were measured using GSA. In total, 3612 optimizations were performed with 301 GSA samples for 12 cases categorized by six locations of the origin (#5, #12, #16, #31, #53, #57) and two leak positions (#13, #15). All cases used five common candidate shelters (#2, #7, #24, #61, #73). The locations of each node in the

Table 4

Scenario definition and simulation results for six test cases.

Case	Input variables Evacuation speed (m/s)	Leak position (-)	Wind speed (m/s)	Wind direction (degree)	Leak rate (kg/s)	Origin (-)	Results Selected shelter (-)	Potential risk (-)	Path distance (m)	Evacuation time (min)
Base	1.7	10	5	15	80	#42	#74	20.15	2144	21.4
Case 1	4	10	5	15	80	#42	#79	20.16	1585	23.8
Case 2	1.7	10	15	15	80	#42	#59	26.19	1055	10.6
Case 3	1.7	10	5	80	80	#42	#79	8.84	1114	11.1
Case 4	1.7	10	5	15	120	#42	#79	20.63	2441	24.4
Case 5	1.7	13	5	15	80	#18	#59	0.01	1052	10.5

**Fig. 5.** Optimal evacuation routes under different input information: (a) optimal paths on the road network map; (b) predicted normalized toxic doses for each case.

geographic map are shown in Fig. 4.

A comparison of the categorized cases in Fig. 6 shows the optimal potential risk distribution of the solutions. The logarithm-scaled sensitivity indices indicate that the most significant factor affecting the optimal potential risk among the surrounding information ν is the wind direction for all cases. Wind speed and leak rate have much smaller influences (0.00–0.04 for wind speed, 0.68–0.86 for wind direction, 0.01–0.21 for leak rate; see Table S2) than wind direction. However, the risk impact based on leak rate changes significantly when a leakage incident occurs at leak position #15. A certain leak rate seems to be required to affect the potential risk of evacuation routes because this area is farther from the residential area than node #13. In addition, leak position #15 has a lower risk distribution than leak position #13, regardless of the origin for a similar reason.

We can infer the inherent influences on the potential risk induced by geographic features from the aspects of the solution distribution lines in Fig. 6. Because the input data samples for GSA are the same for all cases, several GSA solutions at lower risk imply that the geographical features around the given origin and leak position are likely to meet the conditions for safe evacuation. In contrast, a greater number of solutions at higher risk levels contain a higher inherent risk with the given origin and leak position. From this perspective, we determined the number of selections for each shelter and its average potential risk to recognize the inherent risk in the GSA cases, as shown in Fig. 7. The height of the bars

represents the number of selections for each shelter. The colors of the spheres above the bars indicate the average potential risk of all optimal paths heading to the selected shelter. The size of the spheres is the average distance between the optimal paths. Each origin possesses its own preferred shelters; for example, the evacuation routes starting from node #16 mostly select shelter #24 as the destination. The preferred shelter for each origin tends to be located near the origin. However, the frequency of selection was not inversely proportional to the average potential risk. The evacuation paths that departed from origin #57 mostly selected shelter #61; however, shelter #7 had a lower average potential risk and shorter average distance than shelter #61 (see Table S3). This tendency is clearly shown in Fig. 8, where the optimization model selects shelter #61, which exhibits an even higher average potential risk than shelters #2, #7, and #24, regardless of the range of the average potential risk. Because the risk values of these optimal paths are the global minimum, the evacuation routes toward shelter #61 can guarantee safer evacuation even with fatal surrounding variables compared with other candidate paths heading to different shelters.

We investigated the influences of input variables (wind speed, wind direction, leak rate, location of origin, and leak position) on the evacuation route using two indicators: number of selections and an average potential risk value cumulatively imposed on the arrival shelter node. The surrounding information (wind speed, wind direction, and leak rate), especially wind direction, considerably changes the potential risk

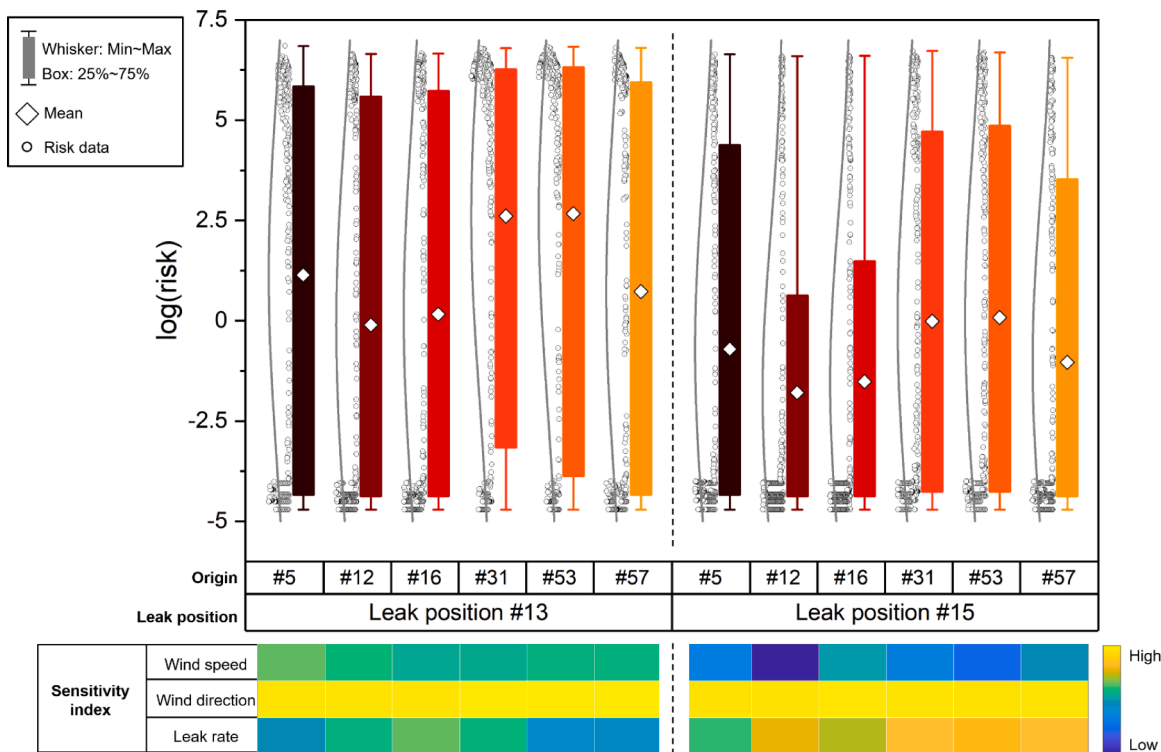


Fig. 6. Sensitivity indices and box plots of accumulated potential risk on optimal paths in global sensitivity analysis samples.

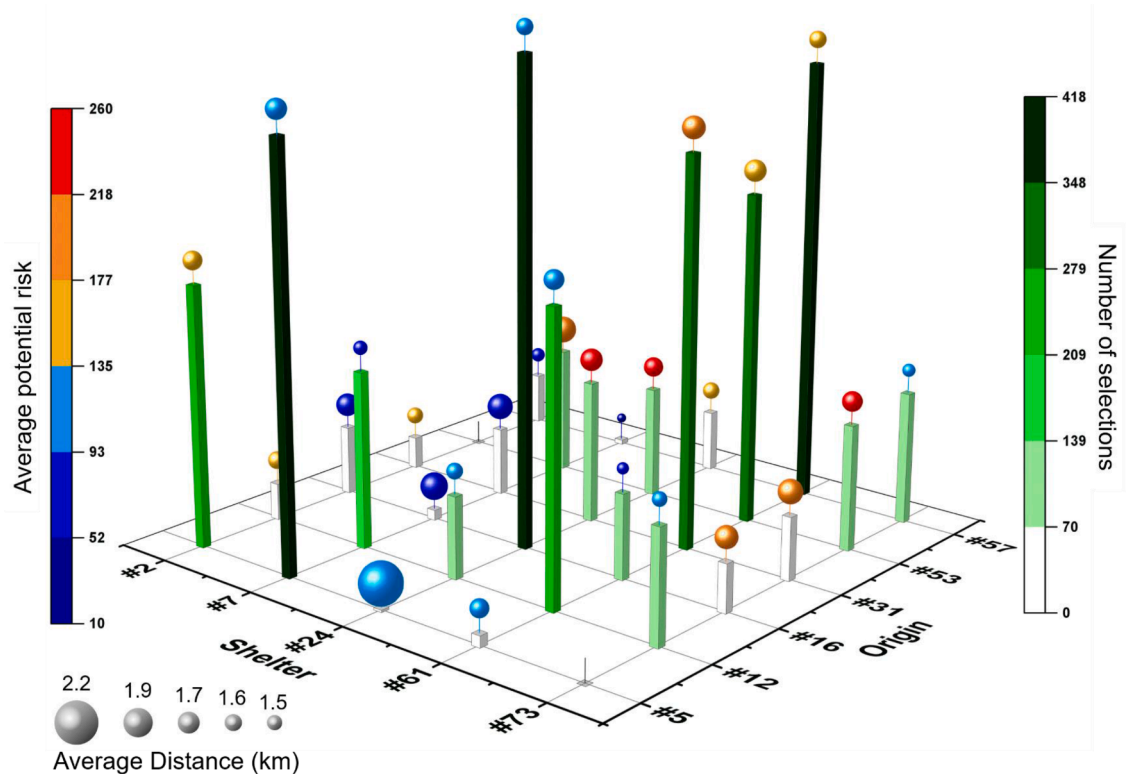


Fig. 7. Average potential risk, traveling distance, and number of selections on optimal paths for global sensitivity analysis samples categorized based on origin and shelter.

of the evacuation route. However, despite the influence of the surrounding information, the location of origin, shelters, and leak position substantially affect a range of potential risks if we consider the average values of potential risks for all scenarios. This is because the geographic

features in the road network and geometric factors extracted from the surrogate model are inherently included in the optimization model. As a result of the GSA, we can also infer that there exists a shelter in the target region that statistically provides low potential risks for the evacuation

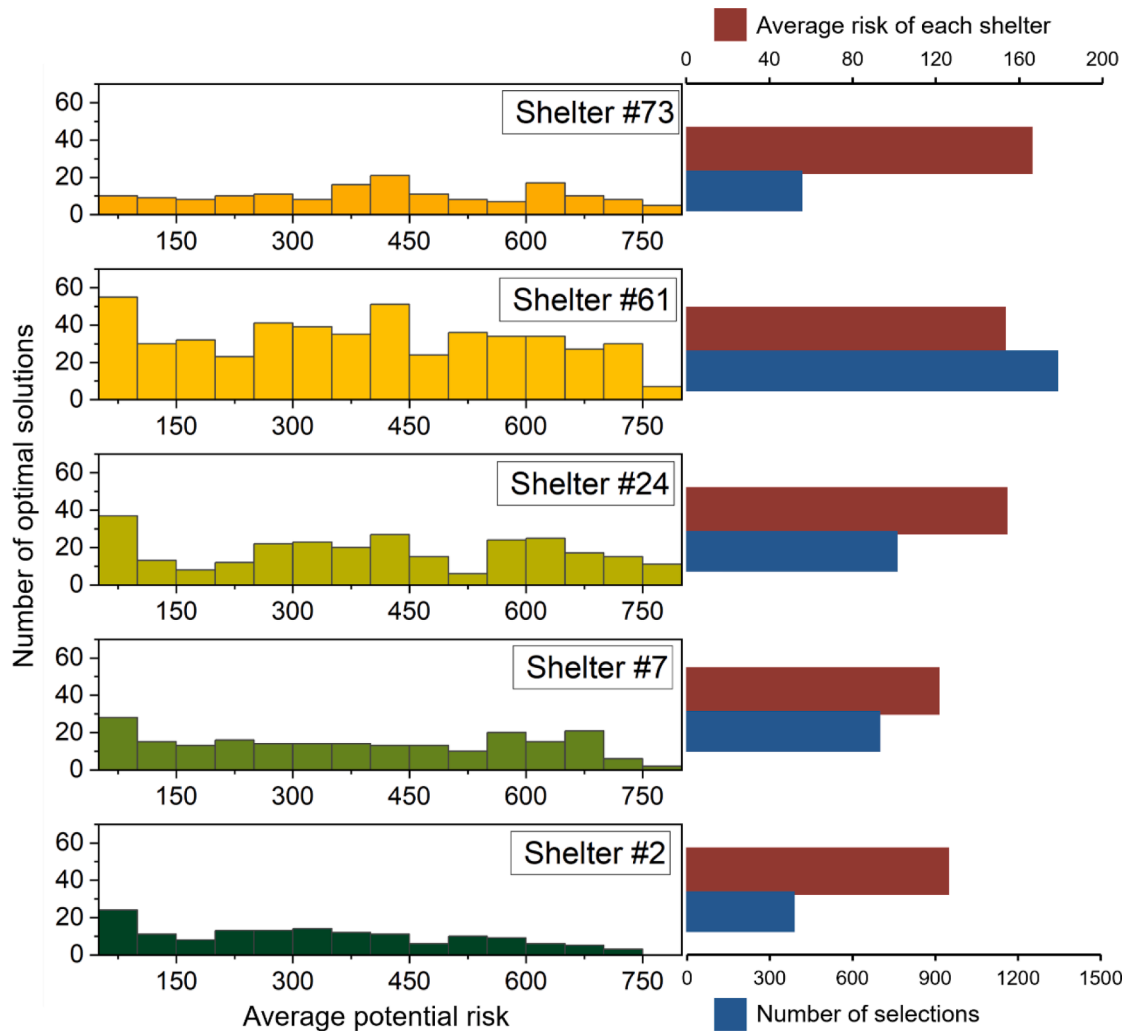


Fig. 8. Distribution of the number of selections and average potential risk of the optimal paths for the global sensitivity analysis samples categorized by each shelter.

route due to the geographic features.

5.3. Analysis of safe shelter location

We performed 5100 CCD samplings to determine the optimal shelter location based on the inference from the GSA results. The average values of the normalized toxic dose obtained from the surrogate model for all simulations are shown in Fig. 9(a). In the heat map, the red-colored region has a higher toxic dose and is distributed around the leak positions, as shown in Fig. 4. The average toxic dose is distributed in the shape of dented ovals and is influenced by the geographical obstacles of mountains and buildings. We totaled the number of selections for each node as the optimal shelter and calculated the average potential risk imposed until arrival at the shelter. Similarly, the average potential risk on each edge was calculated by aggregating the potential risk values on the edges of all solutions. The average potential risks of the nodes and edges were visualized by sizing their relative values, as shown in Fig. 9 (b), Table S4, and Table S5. The thickness of the lines indicates the average potential risk per meter of each edge, and thicker lines tend to be distributed in the peripheral region of the map. The edges with thicker lines seem to traverse a longer distance, are less connected to the other edges, and are surrounded by fewer nodes than those with thinner lines. Conversely, bigger circles, which indicate a higher average potential risk in the node when selected as the optimal shelter, are concentrated around the center of the residential area. This result implies that the smaller circles in Fig. 8(b) can be considered safe

construction sites for evacuation shelters in the case of a toxic gas leak incident, from the perspective of average risk in a uniform distribution of input information ν . Furthermore, the selection frequency of each node is also an important factor in determining shelter positions because nodes with low average potential risk and low selection frequency are not appropriate for shelter construction. From this perspective, the highest frequency node #66 over the entire region might be prioritized as a candidate shelter construction site. However, node #66 is too far from the center of the residential area for evacuation by walking, and its average potential risk is slightly higher than that of some smaller circle nodes. Node #40 may be another suitable construction area for shelter as it is the most selected position in the CCD simulations among the nodes in the residential zone and has a lower average potential risk than node #66. Therefore, we can recommend two specific nodes as safe shelter positions using the proposed evacuation route optimization framework: the node with the highest selection frequency with moderate potential risk and the other near the high population density zone with low potential risk.

We analyzed the number of selections and average potential risk values cumulatively imposed on the arrival shelter node to find an optimal shelter position for evacuation planning. The proposed accelerated optimization framework requires numerous simulations to obtain the data used in the analysis because of its short computation time. In addition, the analyses for the variables of influences and inherent geographic features were derived from surrogate modeling and approximating road networks. Consequently, the proposed system can

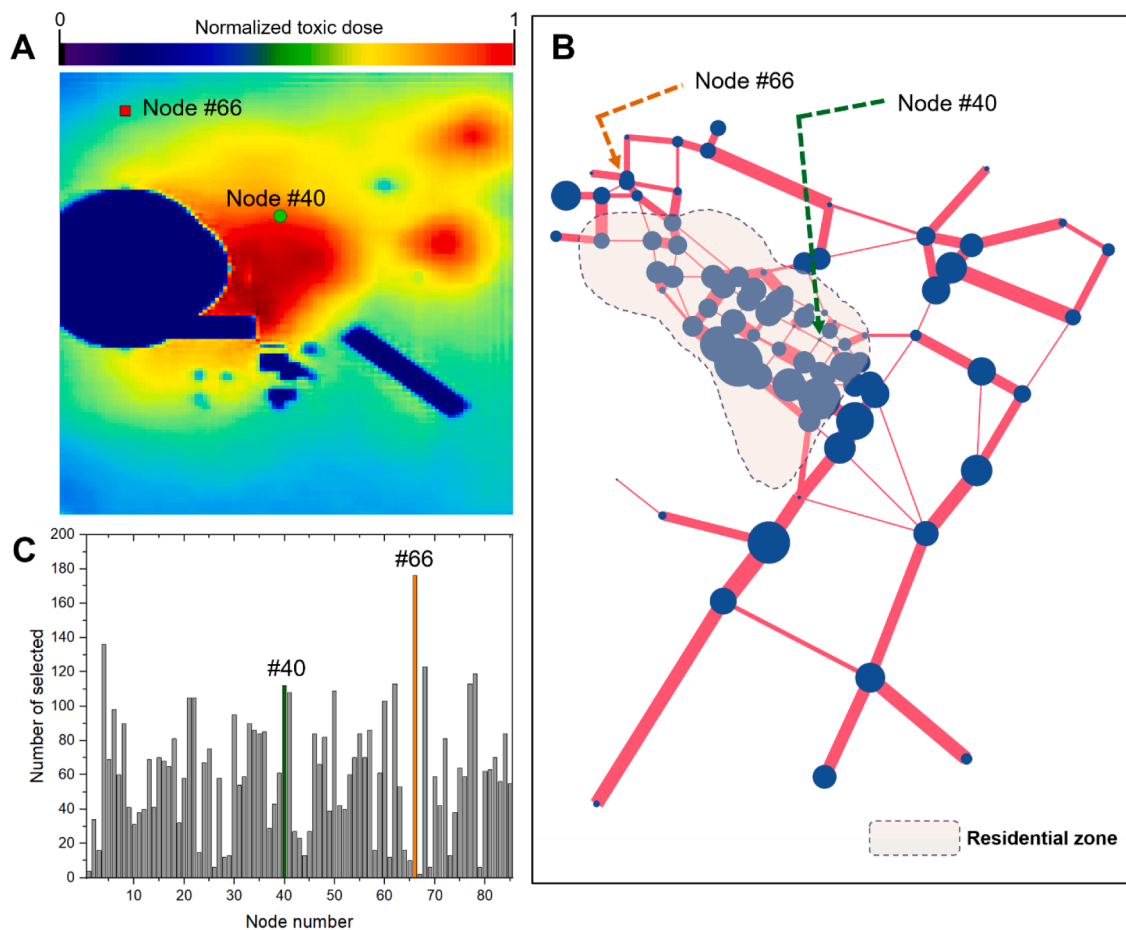


Fig. 9. Results of the central composite design (CCD) for determining safety shelter locations: (a) average normalized toxic dose of all CCD samples; (b) average potential risk of each node and edge for all CCD samples (average potential risk proportional to circle size and line thickness); (c) number of selections for each node as a shelter on the optimal evacuation path.

be used in real-time evacuation planning for individuals and provides insights into the local risk distribution tendency in the target region for safety engineers.

However, several limitations should be addressed in actual incidents. First, solving the MILP optimization problem for an enlarged road network, including small alleys, may require more calculation time and resources. A high-performance computing system should be used to reduce calculation time. Second, the amount of training CFD data is significantly augmented when more leak positions or refined time domains are required. This is because the surrogate models used by the proposed framework are trained differently for each leak position and time domain. However, it did not affect the calculation time for the evacuation route search. We can save training resources and time by examining high-priority spots that contain more lethal gas or obsolete pipelines. Third, we should construct detailed auxiliary decision-making processes not only to show evacuation routes but also to recommend evacuation actions, such as whether to remain in a building or to evacuate. This requires further study on the indoor and outdoor influences of toxic gas dispersion. Despite these limitations, the proposed framework quickly provides the safest evacuation route to individuals in a petrochemical complex and surrounding residential zones, reflecting real-time weather conditions and evacuee location information. The applicability of this study in the real world depends on more sophisticated data collection and the construction of interconnected systems such as safety sensors, information collectors, GIS-based user interfaces, and optimization computers.

6. Conclusion

Evacuation planning for toxic gas leak incidents requires real-time responses for given input variables, such as weather conditions or the characteristics of evacuees. In this study, we defined quantitative risk using a measure of the normalized toxic dose by CFD simulation and constructed a VAEDC-based surrogate model to reduce the computation time required for predicting the normalized toxic dose in the target region. This surrogate modeling approach is combined with the MILP optimization model to enable rapid determination of the safest evacuation route under different scenarios. The proposed framework selects the optimal evacuation path composed of edges defined in the road network, given input variables, including wind speed, wind direction, leakage rate of toxic gas, evacuee's origin, shelter positions, evacuation speed, and leakage position. We conducted a case study on an actual industrial complex that occupies a $2800\text{ m} \times 3800\text{ m}$ area retaining ammonia gas pipelines, to examine the model performance. The computation time for most optimization cases was less than 10 s. The evacuation routes proposed by the model guarantees the lowest potential risk; however, the optimal route detours the shortest path or connects to a shelter farther than the one closest to the evacuee's origin.

Furthermore, the accelerated computation time allows for an additional analysis of the target region. Thousands of simulations were performed using the GSA and CCD methods to apply the optimization model. The main findings of this analysis are presented below.

- From the GSA results, it was found that the most significant factor affecting the risk of an evacuation route is the wind direction because it determines the dispersion direction of the leaked gas.
- In the sensitivity analysis, the distribution of potential risks classified by origin, shelter, and leak positions indicates the inherent influences of the geographic features of those given locations.
- From this perspective, we can plan to construct an optimal evacuation shelter for toxic gas leak incidents by investigating the average potential risk values for every node in the road network and their selection frequency.

To promote the applicability of the proposed method, a greater number of discrete-time domains and leak positions should be considered in the model. A greater number of nodes indicating intersections in the road network and several middle points on a long-distance edge should be added to determine the location of the evacuee. In addition, sampling methods used in GSA and CCD simulations can be substituted by a stochastic approach, such as taking a Gaussian distribution of the leak rate or reflecting meteorological data-based wind speed and direction. Despite these limitations, this study makes two primary contributions.

- We suggested an evacuation route optimization framework based on the VAEDC surrogate model, considering quantitative risks in multiple time domains.
- We significantly reduced the response time required to provide the optimal evacuation route by introducing the surrogate model and the linearized optimization model, while the quantitative risk based on dispersion CFD simulation requires a large computational effort.

CRedit authorship contribution statement

Seung-Kwon Seo: Conceptualization, Visualization, Writing – original draft. **Young-Gak Yoon:** Data curation, Visualization. **Ju-sung Lee:** Data curation, Visualization. **Jonggeol Na:** Conceptualization, Methodology, Supervision. **Chul-Jin Lee:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

Funding: This work was supported by the National Research Foundation of Korea (NRF) Grant funded by the Korean Government (MSIT) (NRF-2021R1C1C1012031), and the Chung-Ang University Graduate Research Scholarship in 2019.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.res.2021.108102](https://doi.org/10.1016/j.res.2021.108102).

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